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XML and DataBase

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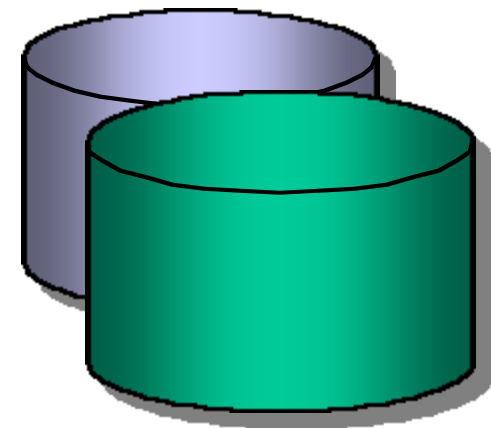
<http://java.sun.com/xml>

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- **Advantages of XML and DB**
- **Related J2EE**
- **XML format from Database**
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- **Fetch and Update Database**

Overview

- **Database**
 - a back-end server for Web Applications
 - CGI
 - JDBC
 - Embedded SQL



Overview

- XML
 - eXtensible Markup Language
 - **Target:**
 - If it is be represented with **Standard open format** regardless of any data, **Freely Give and Take!**
 - Structure and Semantic
 - Hierarchy Structured
 - self-describing

Overview

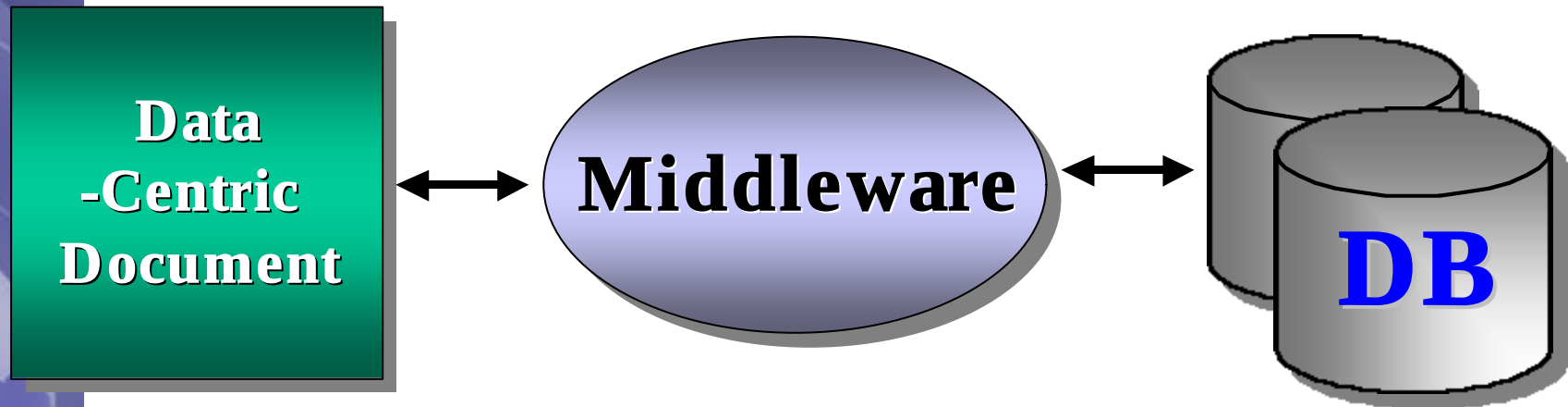
- **XML**
 - **Optionally Driven DTD**
 - **DTD**
 - **Extract, understand, analysis XML document content**
 - **Define XML Data Structure**
 - **Efficient Storage**
 - **Formulation**
 - **Optimization**
 - **Processing**

XML and DB Mapping

- **Data-Centric document**
 - **a fixed structure**
 - a dining room menu
 - an airplane schedule table
 - Requested order

XML and DB Mapping

- **Data-Centric document**



XML and DB Mapping

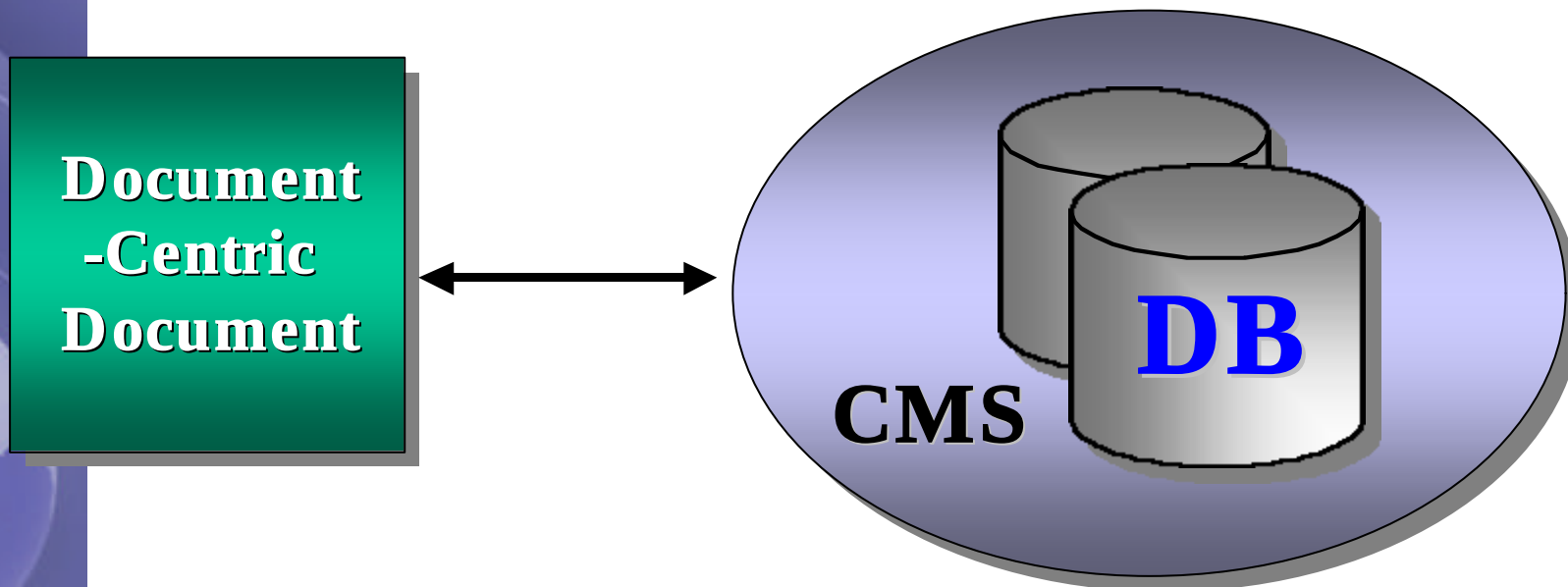
- **Data-Centric document**
 - **Template-Driven**
 - Mapping **by Template Defined Previously**
(Only RDB)
 - **Model-Driven**
 - Mapping **by DB structure**
 - RDB: Table model(single table)
 - OODB: data-specific object model(class)

XML and DB Mapping

- **Document-Centric document**
 - **Non-fixed structure**
 - **Important Sibling order With Mixed content**
 - **Book**
 - **Email**
 - **Promotion**
 - **Mostly HTML, XHTML document**

XML and DB Mapping

- **Document-Centric document**



XML and DB Mapping

- **Document-Centric document**
 - **CMS(Content Management System)**
 - **Out of the Box Use**
 - XML Document Put In **One Column**
 - **Writing Your Own System**
 - **several table create** accordingly **DOM structure**

XML and DB Mapping

- **Mapping Problems with RDB**
 - **Destroy Integrity**
 - **Low Scalability(Require Join)**
 - **Narrow Link and Navigation**
 - **Restrictively Indexing**

Advantages of XML and DB

- **Representation Structured Data Without Additional Information**
 - DB: **Schema**
 - XML: **DTD, Schema**

Advantages of XML and DB

- **Development XML Applications Upon Relational DB**
 - **For OLTP/OLAP**
 - **Applications With The existing RDBMS**
 - **high expenses New XML DBMS**

Advantages of XML and DB

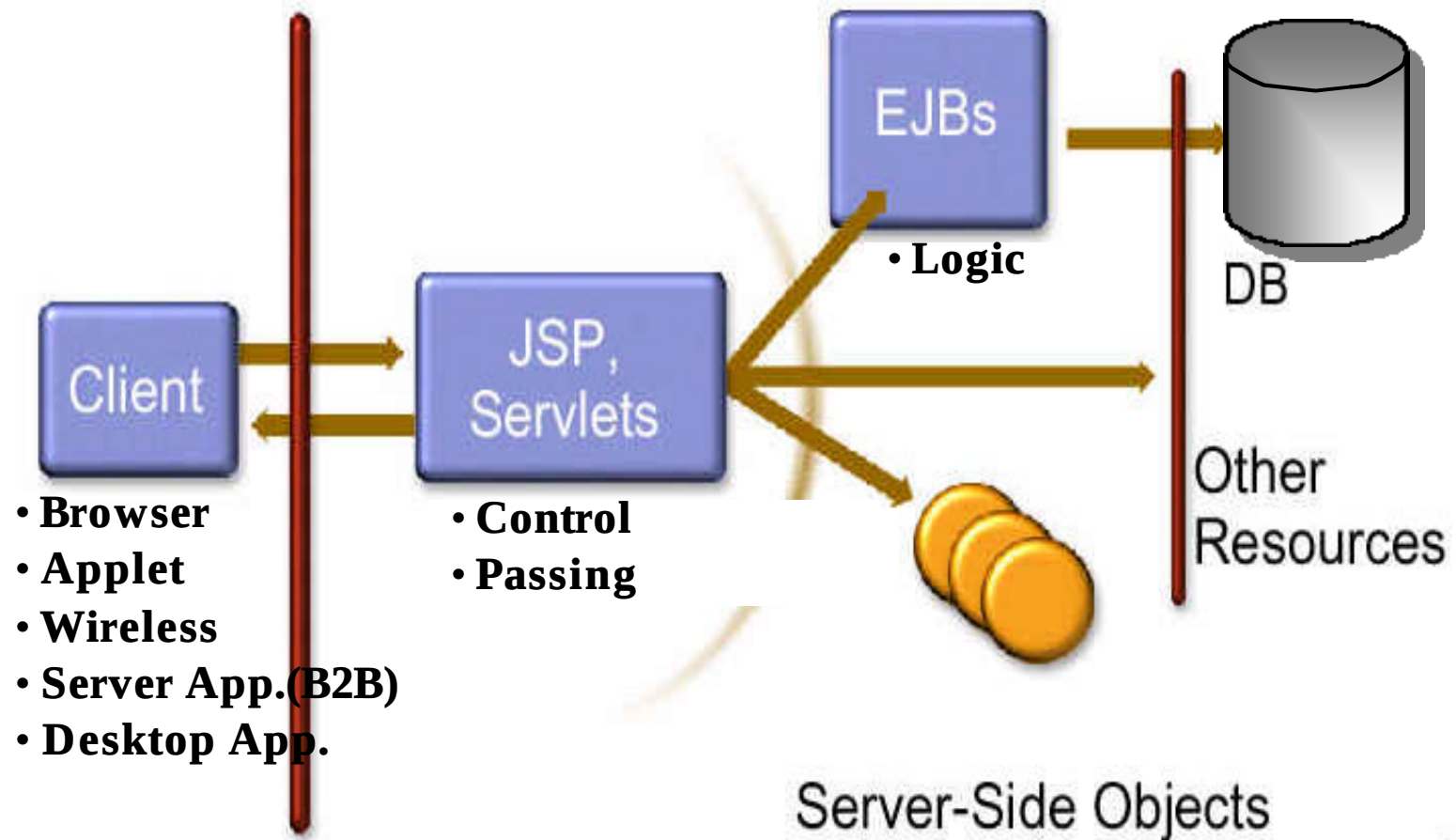
- Transform Various Client's Environment document format



Related J2EE

- **J2EE and XML**
 - **Combining XML**
 - **Generating XML**
 - **Transforming XML**

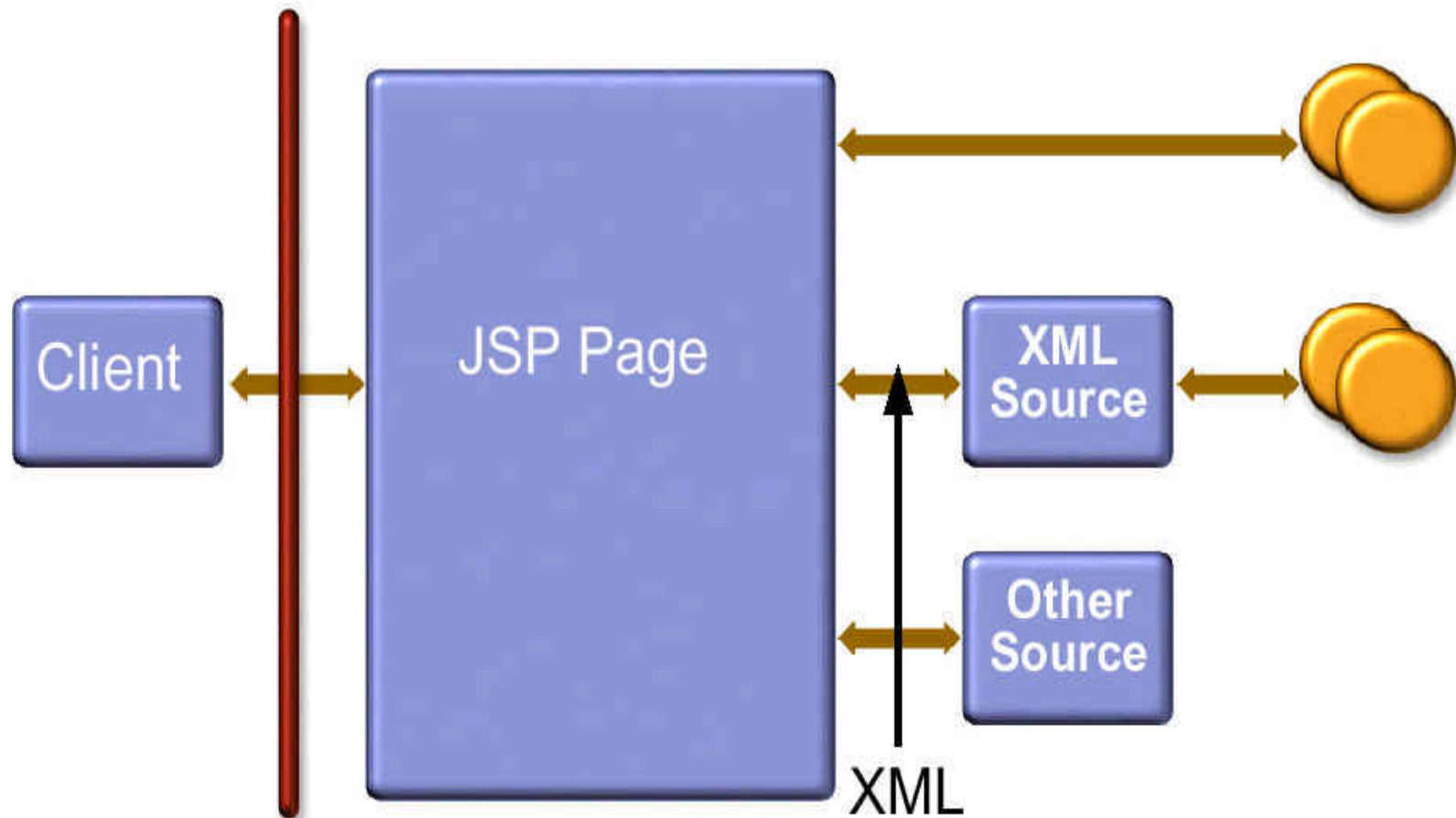
J2EE and XML



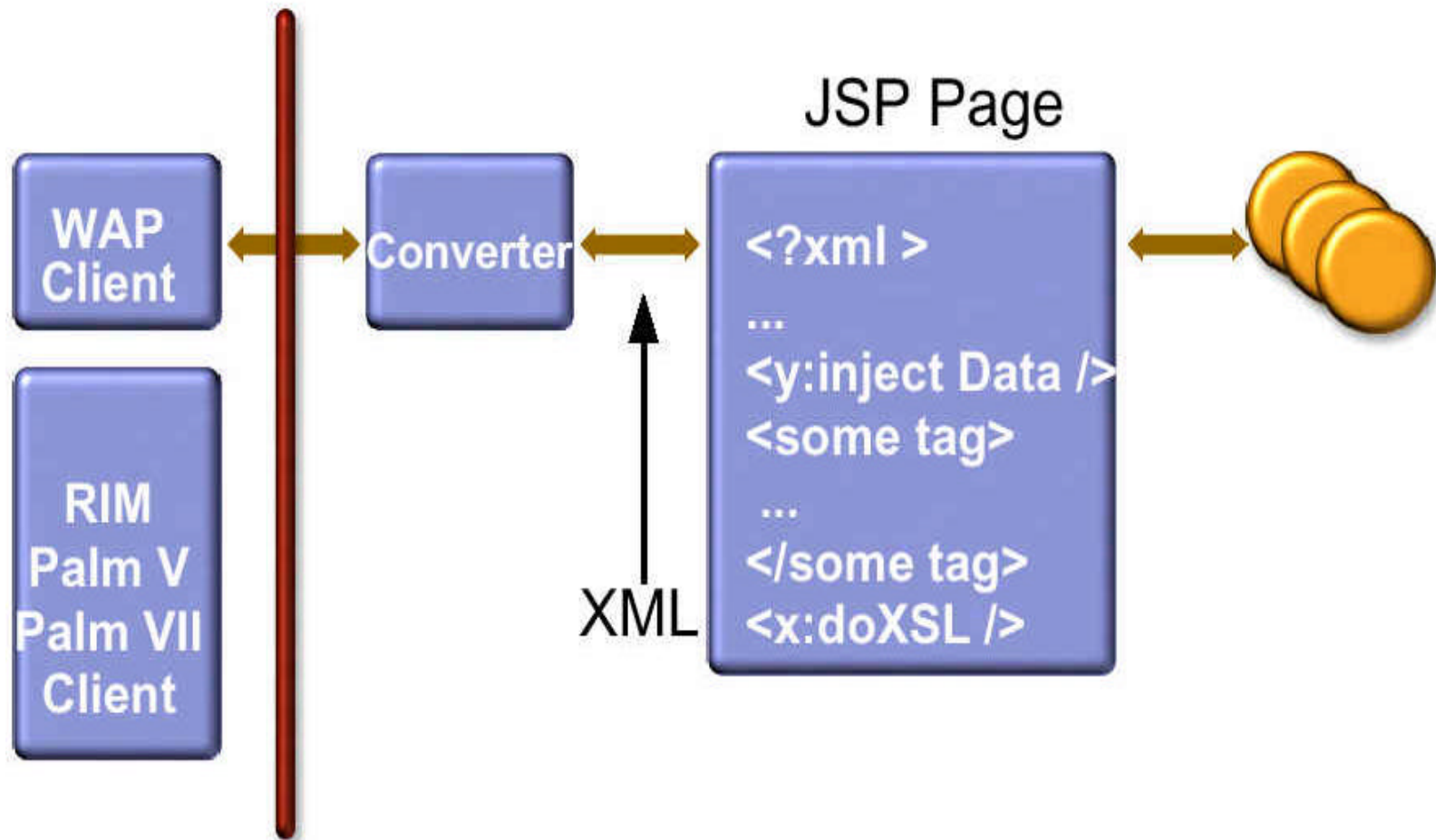
J2EE and XML

- **Combining XML**
- **Generating XML**
- **Transforming XML**

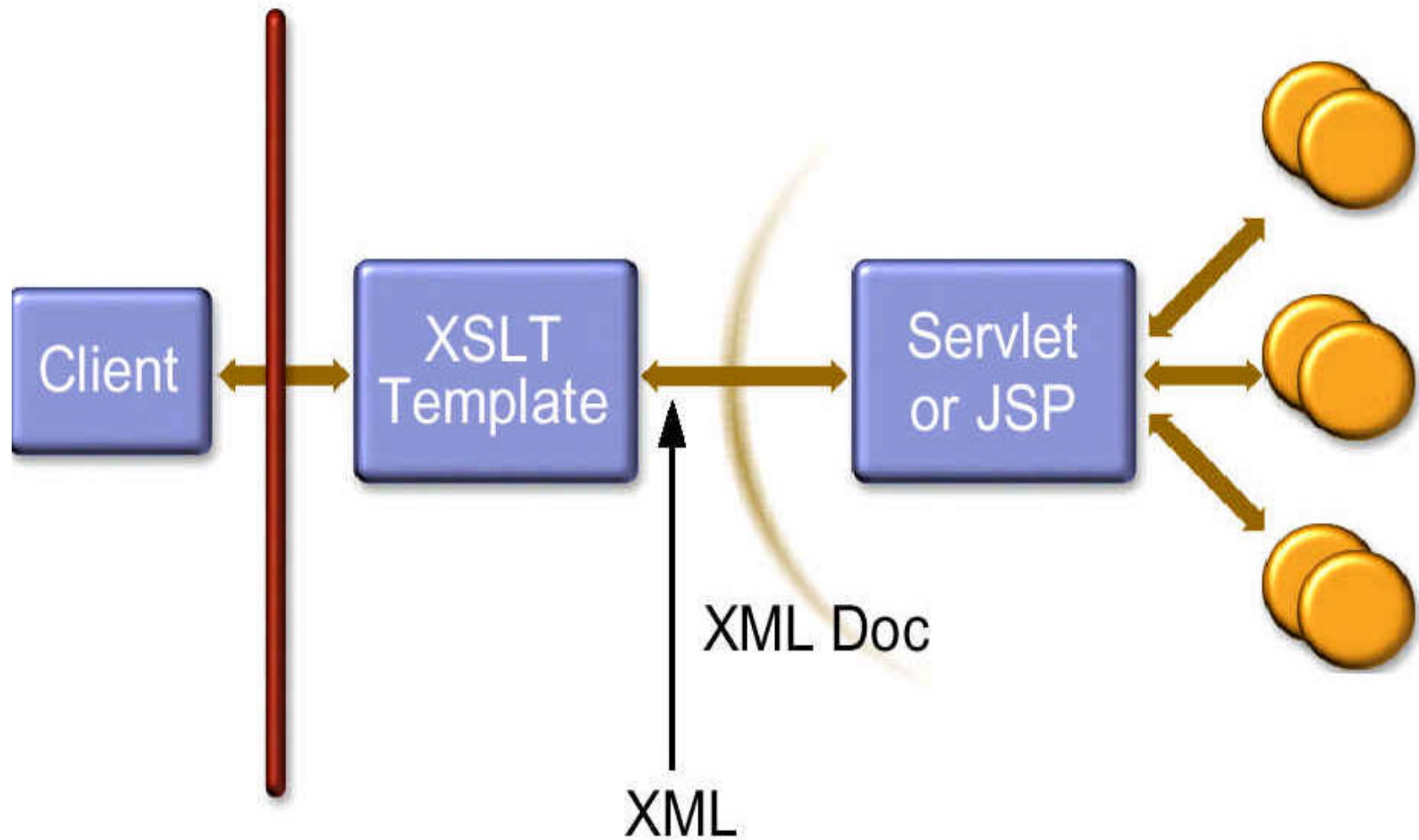
COMBINING XML



GENERATING XML

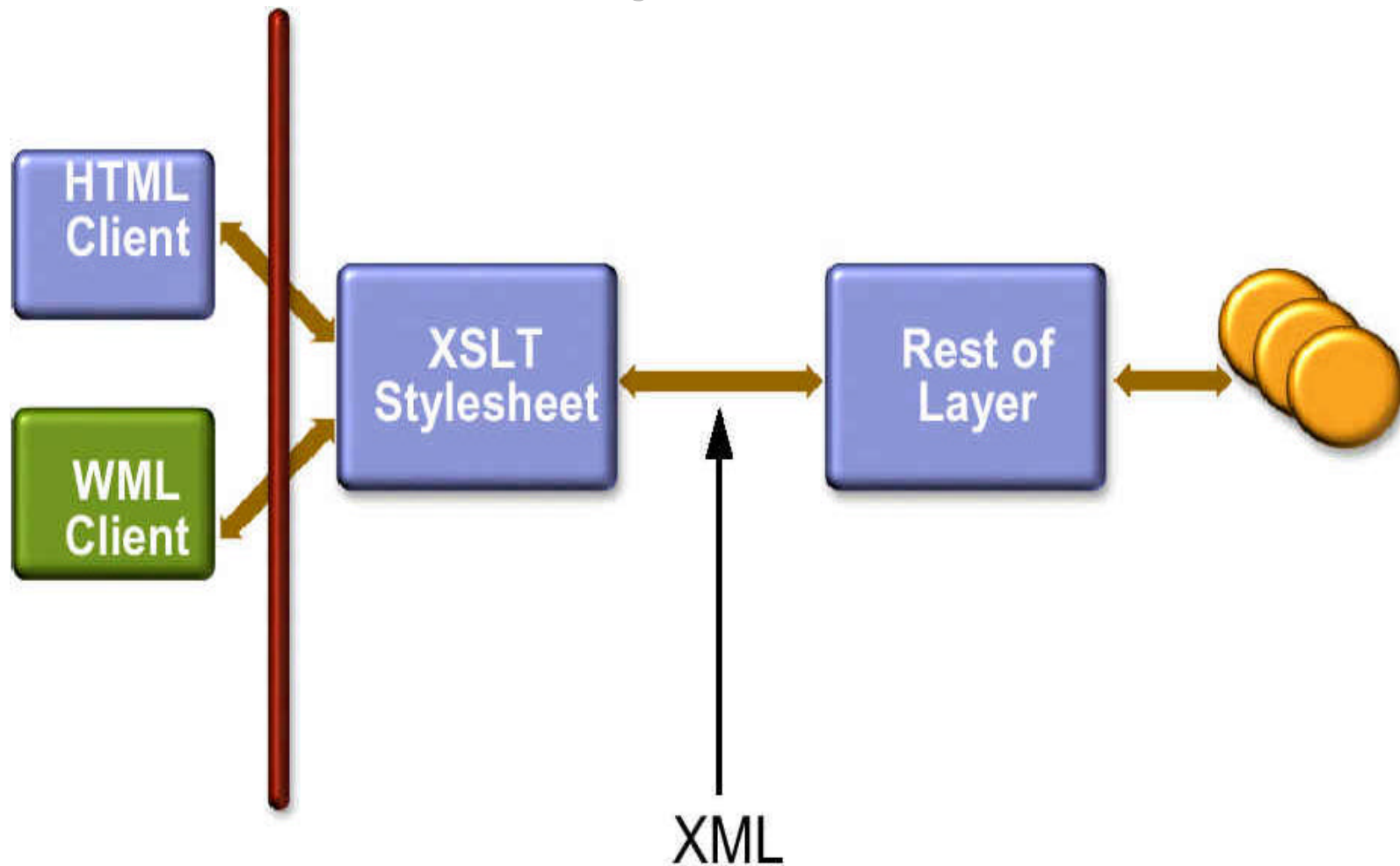


TRANSFORMING XML



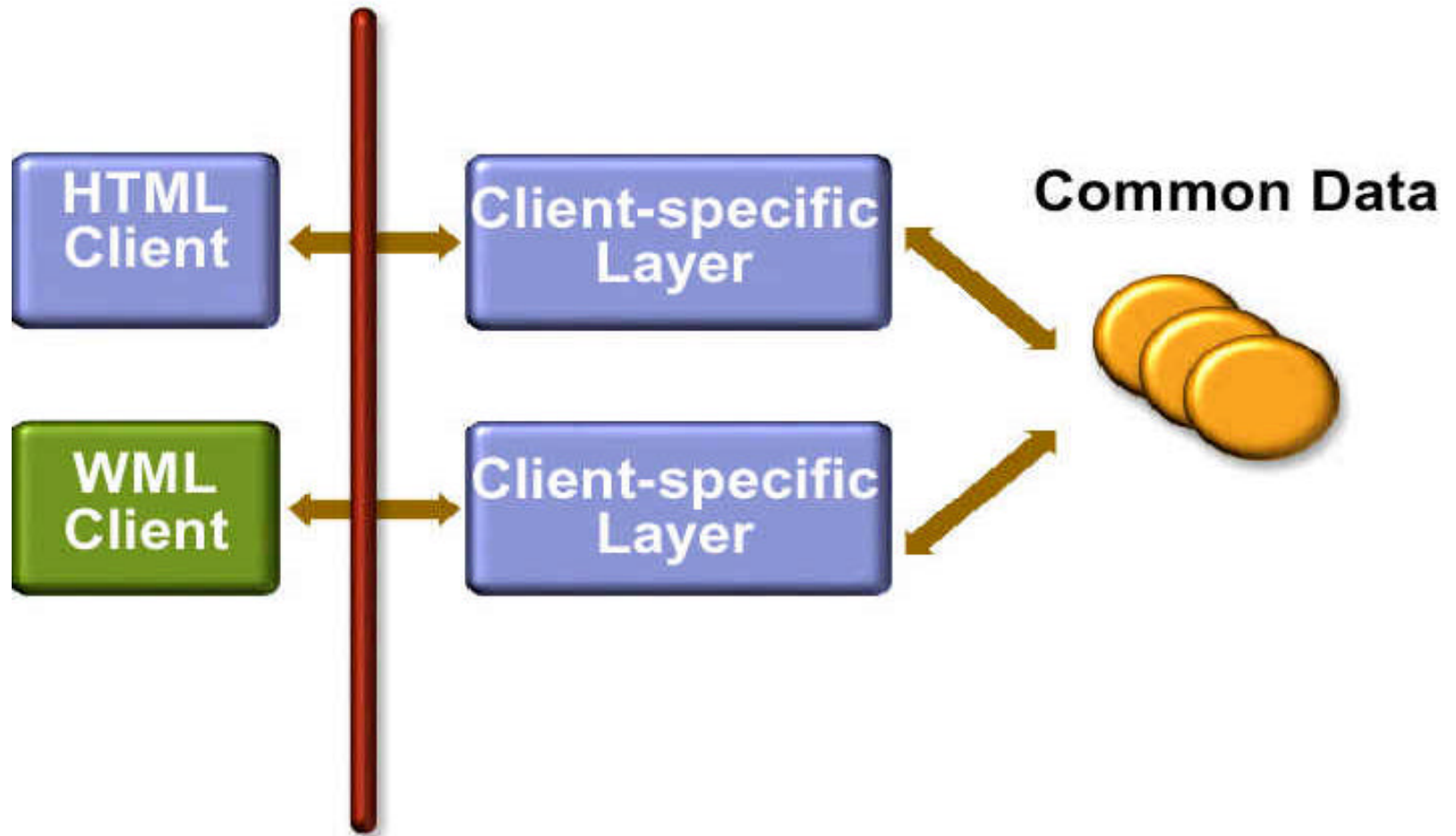
TRANSFORMING XML

►► XSLT StyleSheet



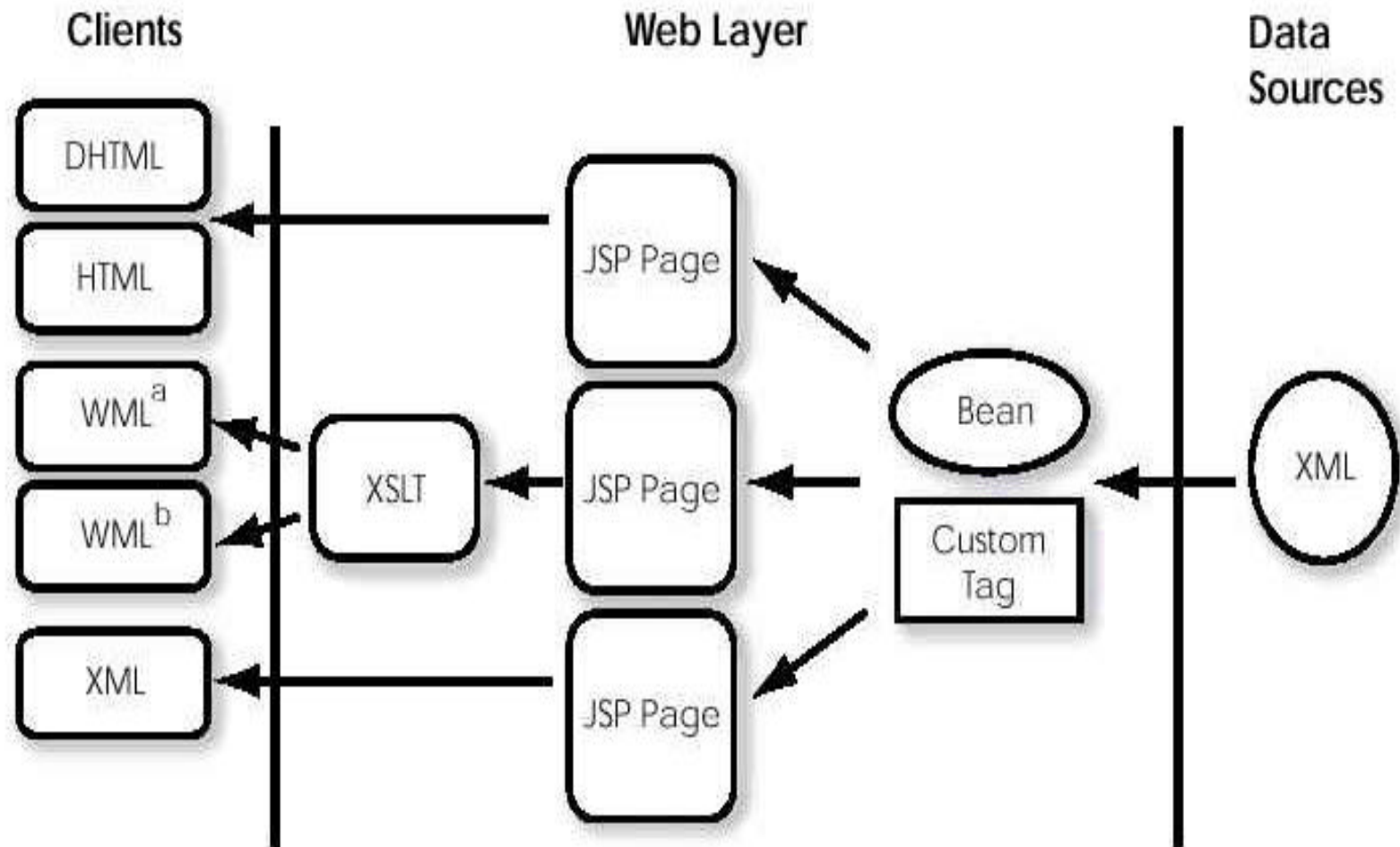
TRANSFORMING XML

► Short Pipelines



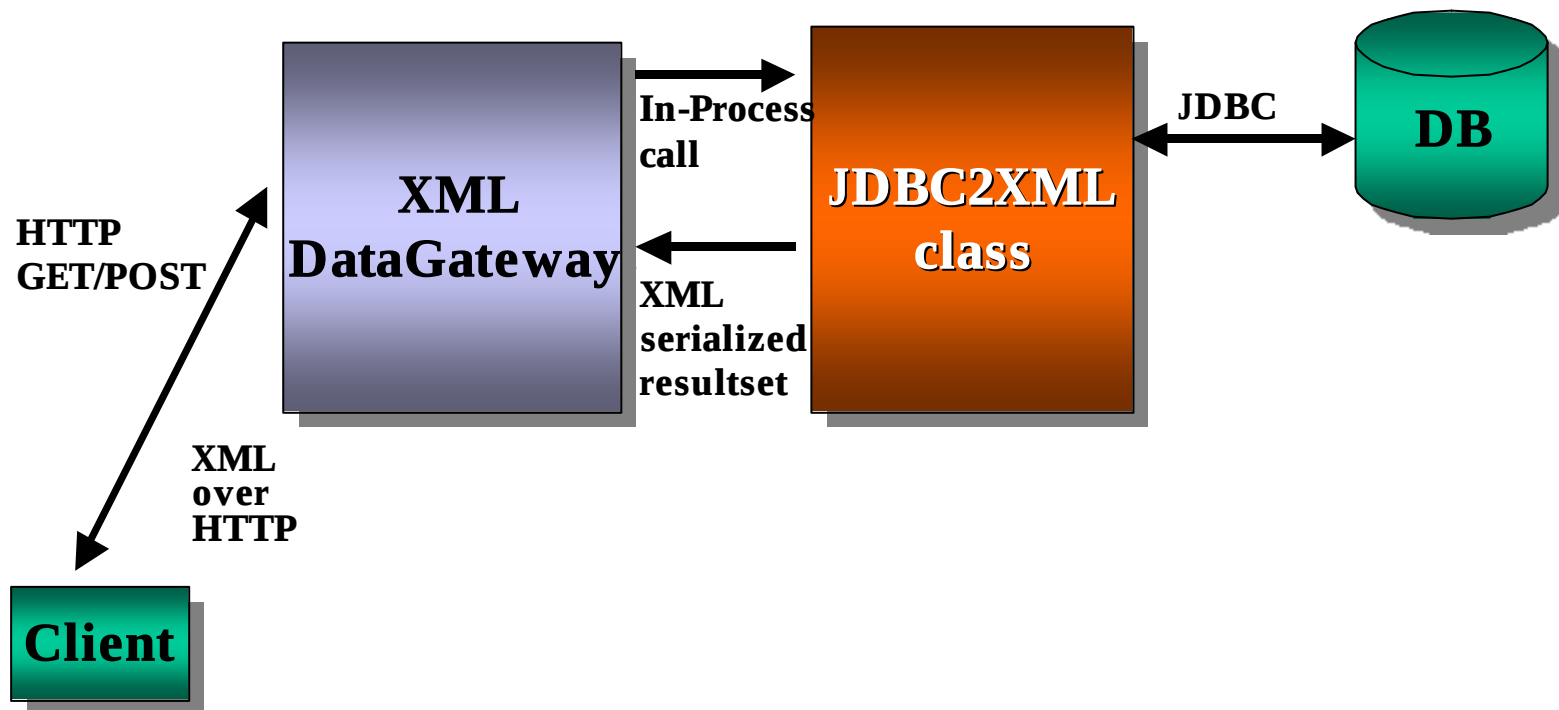
TRANSFORMING XML

►► Combined Approach



XML format from Database

- XML format ResultSet from Database

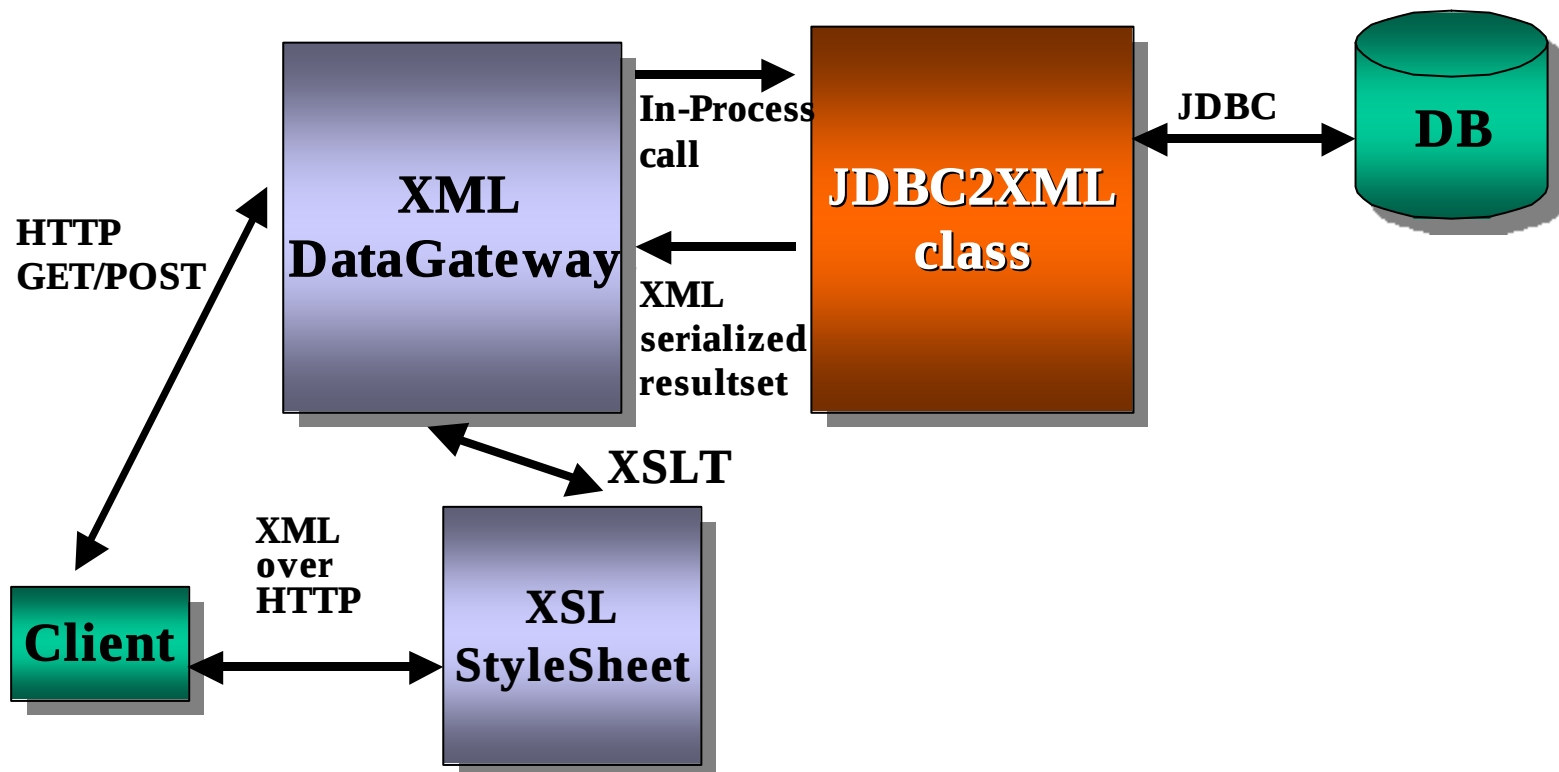


XML format from Database

- **XML format ResultSet from Database**
 - **JDBC2XML class**
 - **encodeXML Method**
 - **Serializing the Result Set as XML**
 - **Executing the Query**
 - **XMLDataGateway Servlet**

HTML format from Database

- **HTML format ResultSet from Database**

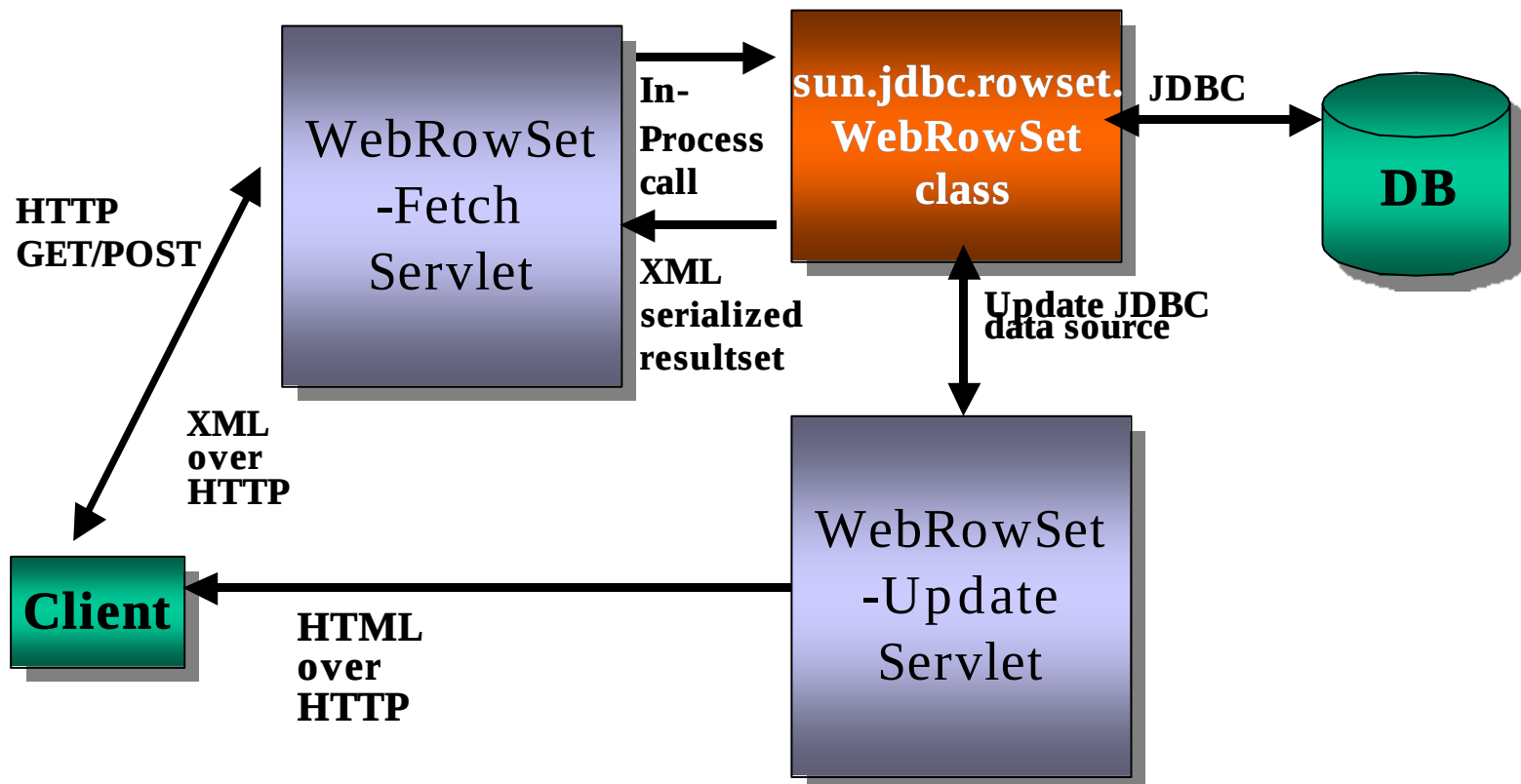


HTML format from Database

- **HTML format ResultSet from Database**
 - **JDBC2HTML Servlet**
 - **XSL stylesheet transform**
 - **query database**

Notify Fetch and Update Database

- Notify Fetch and Update Database



Notify Fetch and Update Database

- **Notify Fetch and Update Database**
 - **sun.jdbc.rowset.WebRowSet class**
 - **WebRowSetFetchServlet class**
 - **WebRowSetUpdateServlet class**
 - **WebRowSetHTTPClient class**

Thanks

